

$$f(x) = x^{\ln x}$$

$$f(x) = (e^{\ln x})^{\ln x}$$

$$f(x) = e^{\ln^2 x}$$

$$f'(x) = \frac{2e^{\ln^2 x} \ln x}{x}$$

$$f'(6) = \frac{e^{\ln^2 6} \ln 6}{3}$$